

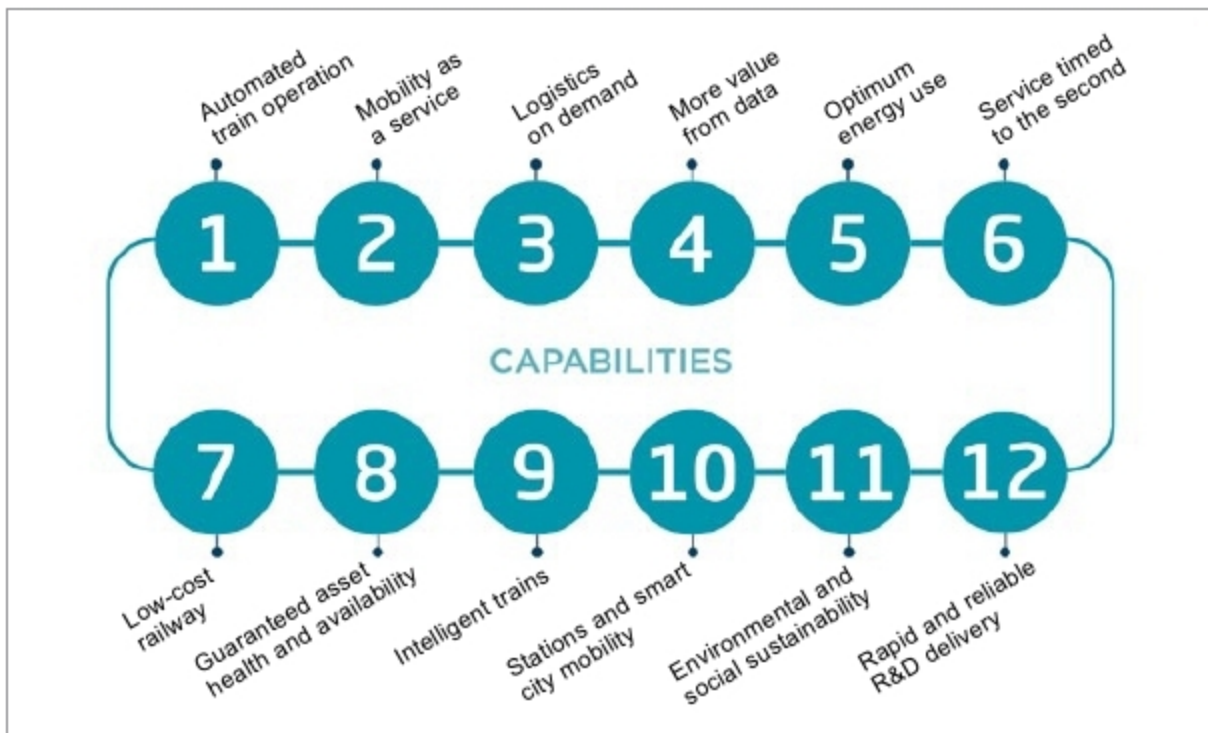


Fig. 1: Twelve capabilities required for the realisation of Shift2Rail vision (Credit: <https://shift2rail.org>)

and secure customer experience.

Transportation systems are complex with respect to technology and operations due to the involvement of a wide range of human actors, organisations and technical solutions. There is a need to apply intelligent computerised systems for the operation and control of such complex environments, such as computerised traffic control systems for coordinating airline transportation.

Industry 4.0 is enabled by smart systems and Internet-based solutions. Maintenance is one of the application areas of self-learning, and smart systems can predict failure and trigger maintenance by making use of the Internet of things (IoT).

Digital railway solution

The digital railway programme is focussed mainly on digital signalling technology, which aims to enhance safety and speed up train movement in a congested network. If all data from signalling, rolling stock and passenger traffic control systems is brought together on a common platform, the entire network will be able to communicate seamlessly and instantaneously. The key to digitisation is interoperability of systems while retaining a critical approach to data security.

Rail service information could even be integrated with other transport modes, such as bus and taxi services, to guide passengers through

smooth door-to-door journeys. Holistic data management could lead to the transformational change in real-time intelligent traffic management and in-cab signalling. This could improve customer satisfaction, with station information systems and personalised messaging providing passengers with all the relevant information they need.

Digitisation of railways includes:

- B-scan ultrasonic rail flaw detection (both non-stop and stop-and-verify systems) and track inspection with automated high-speed test trains
- Train control system levels 2 and 3 for high-density routes to increase network capacity and maintain the required safety standards
- Increased surveillance of personnel with both interior and exterior locomotive-mounted video surveillance to improve monitoring
- Track-laying machines for mechanisation of construction
- Electrification through machines such as self-propelled overhead electrification laying trains
- Complete train scanners for improved diagnostics and maintenance
- Use of distributed power to improve the efficiency of train operations with coordinated acceleration and deceleration
- Establishment of smart railway stations by implementing access control at entry points
- e-ticketing with services such as

infotainment and app-based systems

- Use of training simulators and virtual reality (VR) training systems to improve personnel capabilities

Digital twin—the digital platform for railways and airways

A digital twin refers to a virtual replica of a physical asset, like an aircraft engine or a rail engine. It is a vital element of the digital rail solution that is continually updated as per the rail network. It enables engineers to test detailed what-if scenarios that could help in decision-making around the planning of enhancement and maintenance programmes. It could identify the most-valued solution that would have the greatest efficiencies and minimise disruptions.

Every year unplanned maintenance, flight delays and cancellations cost millions of dollars to every airline. Optimisation of operational availability can reduce operating costs due to maintenance. Monitoring engines in use with a digital twin can help airlines achieve this. A digital twin can display how the engine is running to engineers on the ground, while the aircraft is still in the air. Later, it can be linked to IT systems to help streamline and optimise maintenance processes and operational availability.

For each asset, engineers compile thousands of data points specifically during the design and manufacturing phases of the engine. These are then used to build a digital model that tracks and monitors an asset in real time, providing essential information throughout an asset's lifecycle, such as engine temperature, pressure and airflow rate.

These virtual models of the assets can provide early warnings, predictions and even a plan of action by simulating what-if conditions based on weather, performance, operations and other variables. This way these models help keep the aircraft in service for longer.

Role of sensors in predictive maintenance

Sensors are constantly getting cheaper. This will increase their implementation as a leading real-time data pro-